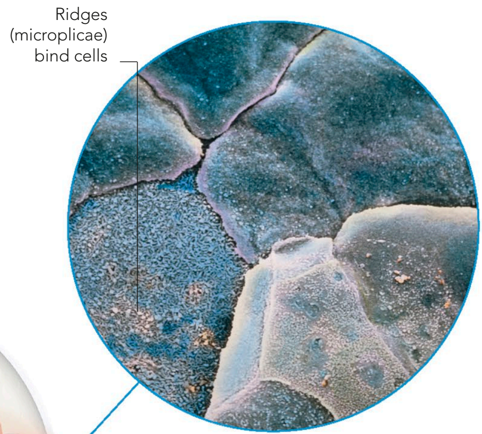
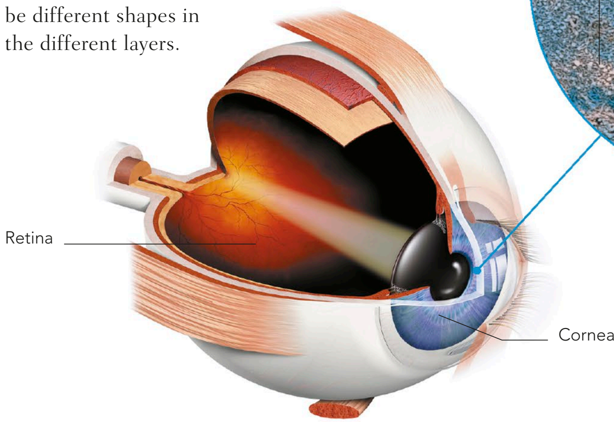
SIMPLE AND LAYERED EPITHELIUM

Simple epithelium is composed of a single layer of cells. This type of tissue is often found in areas where substances need to pass through easily, a single-cell thickness offering minimal resistance. For example, in the air sacs in the lungs, simple epithelium allows the exchange of gases to take place. Layered (stratified) epithelium has two or more layers, and is better for protection in areas such as the mouth or oesophagus. Some complex epithelium has more than five layers, but two or three is more usual. The cells may be different shapes in the different layers.

CORNEA STRUCTURE

The epithelium covering the cornea is transparent and about five layers thick. It permits light rays to enter the eye.





EPITHELIUM IN THE EYE

The eye contains two types of epithelium: simple epithelium in the pigmented layer of the retina, and stratified squamous epithelium in the domed front "window" of the cornea.

TYPES OF EPITHELIAL CELL

The cells that make up the epithelial layers are usually classified according to their shape. Since most epithelial cells, as a consequence of their locations in the body, are subject to friction, compression, and similar physical wear and tear, they divide rapidly to replace themselves.

Squamous
Plate-like or flattened cells, wider than deep, resembling paving slabs or crazy-paving; flattened nucleus.

Features: Cells allow selective diffusion, or permeability, allowing certain substances to pass, owing to thinness of the layer.

Cuboidal
Cube- or box-shaped cells, occasionally hexagonal or polygonal; nucleus usually in cell centre.

Features: Substances absorbed from one side of the layer can be altered as they pass through the cytoplasm of the cuboidal cells, before leaving.

Columnar
Tall, slim cells, often square, rectangular, or polygonal; large, oval nucleus near cell base.

Features: Protect and separate other tissues; may be topped with cilia for movement of fluid outside the cell or microvilli for absorption.

Glandular
Epithelial cells modified for secretion, usually cuboidal or columnar with secretory granules or vacuoles.

Features: Layers of these cells may be infolded to form pits, pockets, grooves, or ducts, as in sweat glands.